Eslam Usama Mahmoud

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Professional Experience

Zikola *Mansoura, Egypt*
AI Engineer *April 2026 – Present*

- **AI Shopping Assistant (WhatsApp + Web):** Built a bilingual (Arabic/English) assistant for online stores that answers customers from each store's real catalog using LangGraph, LangChain, and RAG. Handles voice notes, photos, order tracking, and human handoff, serving many stores from one platform.
- **Real-Time Audio & Meeting Intelligence Engine:** Built a Chrome extension that records and transcribes meetings and attributes speech to the right attendees, with an asynchronous LLM pipeline that turns multi-hour transcripts into validated action items, summaries, and sentiment reports. Secured with passwordless WebAuthn authentication, multi-tenant RBAC, and audit logging.
- **Salla App for AI Product Recommendations (Rashihli):** Built and shipped a multi-tenant Salla app that adds an Arabic-first shopping quiz to storefronts, generated from each store's own catalogue, with subscription plans and a merchant panel inside the Salla dashboard. Demo video
- **Arabic Voice AI (Concierge + Order-Confirmation Calls):** Built a real-time Arabic voice assistant (Deepgram, ElevenLabs) with barge-in and human handoff, plus an event-driven calling engine on LangGraph and Vapi that confirms cash-on-delivery orders by phone and updates the order, with compliance rules enforced in code.
- **Multi-Platform Media Analytics & Automation:** Developed an automated media-buying ETL engine processing multi-platform CSVs (Meta, TikTok, Snapchat) with auto-calculated KPIs (ROAS, CPA, CTR, CPC) integrated into n8n AI-agent workflows.

Projects

Volleyball Group Activity Recognition Case study | GitHub | Live Demo

- Reproduced and extended the CVPR 2016 hierarchical deep temporal model in PyTorch on 4,830 clips: built an 8-baseline ablation ladder (ResNet50, per-player LSTM, team-separated pooling) reaching **88.8%** group-activity accuracy (paper: 81.9%) and a joint model with 75.7% player-action accuracy. Fine-tuned YOLOv8n (0.98 mAP50) with ByteTrack to turn raw match video into a timestamped event timeline, served through a FastAPI backend and web studio.

Elara – AI-Powered Educational Ecosystem *Multi-Module System*

- **Elara TTS (Egyptian Arabic Engine)** GitHub
  Curated a 14-hr Egyptian speech dataset (~18.9k samples, VAD via faster-whisper) with custom NLP normalization; fine-tuned Spark-TTS (0.5B) via Unsloth and Wav2Vec2/BiCodec tokenization for educational speech.
- **LLM Cognitive Evaluation Engine** GitHub
  Fine-tuned Qwen2.5-14B via QLoRA (4-bit NF4) on 3,946 sessions; deployed on Modal with vLLM + guided JSON decoding to auto-generate schema-validated reports on student misconceptions.
- **RAG Learning Assistant** GitHub
  Built an ingestion pipeline with recursive chunking and FAISS vector retrieval to ground LLM responses in course materials with precise source citations.
- **Student KPIs & Predictive Analytics** GitHub
  Developed end-to-end ML pipelines using gradient-boosted ensembles to forecast academic performance and feed real-time instructor monitoring dashboards.

Akhessai – AI-Assisted Special Education Assessment Platform GitHub

- Built an Arabic-first (RTL) multi-role platform (React/TypeScript, FastAPI, PostgreSQL) where a deterministic YAML rule engine scores 17 developmental domains and triages 6 clinical specialties; RAG over a 120-document Arabic clinical corpus (multilingual MiniLM embeddings, Qdrant) grounds a Gemini 2.5 Flash report generator constrained to a Pydantic-validated JSON schema, with every draft routed to a licensed specialist for sign-off; JWT-based RBAC and Argon2 password hashing.

LUNA16 3D Pulmonary Nodule Detection System GitHub

- Built a 3-stage 3D pipeline on 888 CT scans with classical 7-step lung segmentation in NumPy/SciPy (HU thresholding, morphological closing) and trained DenseNet-121 from scratch for high-recall candidate detection and false positive reduction.

Arabic Medical LLM Fine-Tuning & Deployment GitHub

- Fine-tuned Qwen2.5-1.5B via LoRA/Unsloth on merged Arabic medical QA datasets; deployed an async CUDA inference FastAPI backend with responsive chat UI.

Building Energy Consumption Digital Twin GitHub

- Hierarchical regression system combining XGBoost sub-models with a Neural Network meta-learner, served via Flask REST API with real-time scenario simulation.

Technical Skills

Programming	Python, C, C++, PHP, JavaScript, TypeScript
AI & ML Frameworks	PyTorch, TensorFlow, Scikit-learn, Keras, XGBoost, LangGraph, LangChain
Data Science	Pandas, NumPy, Matplotlib, Seaborn, OpenCV
Computer Vision	Object Detection & Tracking (YOLOv8, ByteTrack), Action Recognition, 3D Medical Imaging
NLP & Speech	Hugging Face Transformers, PEFT/TRL, NLTK, Wav2Vec2, Faster-Whisper
MLOps & Deployment	Docker, FastAPI, Flask, Streamlit, vLLM, Unsloth, Vercel Serverless
Web & Integration	React, MV3, n8n, Google Apps Script, WebAuthn, Webhooks
Databases & Tools	PostgreSQL, SQL Server, SQLAlchemy, Git, GitHub, Jupyter Notebook
Languages	Arabic (Native), English (Fluent)

Education

Bachelor of Computer Science *2022 – 2026*

Faculty of Computers and Information, Mansoura University, Egypt *Graduated*

Certifications & Training

- **Machine Learning Diploma** – CSkilled Academy *130+ hrs, Sep 2026*
  Stanford-inspired curriculum covering regression, regularization, model selection, classification metrics, end-to-end ML, unsupervised learning, and deep learning with PyTorch (CNNs, RNNs). Certificate
- **Data Analysis & Freelancing** – ITIDA & NTI *120 hrs, Aug–Sep 2024*
  Covered data analytics workflow, freelancing dynamics, proposal writing, and project pricing strategies.
- **Data Science & Web Development** – MCIT *120 hrs, Jul–Aug 2025*
  Hands-on training in core data science concepts, business domain modeling, and web development.